

ARCTOS CONSORTIUM • UC DAVIS MSBA

Arctos Querybot

User Guide

Natural-language search for the Arctos Museum Database
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What is Arctos Querybot?

Arctos Querybot is a natural-language search interface for the Arctos Museum Database. Type a plain-English question, and the app generates a structured Arctos search URL — no manual form-filling required.

— Type a plain-English question → get a ready-to-use Arctos search URL	— Results link directly to live specimen records in the Arctos database
— The app recognises institutions, collections, and GUID prefixes automatically	— No login or API-key setup required — open the URL and start searching
— Fields the app doesn't recognise (taxa, dates, geography) are handled by Gemini AI	— Submit optional feedback after each result to help improve accuracy over time

Accessing the app

The app is hosted on Streamlit Community Cloud.
No installation or account setup is needed.

TO ACCESS

Open your browser and navigate to the app URL provided by your administrator.

The Gemini API key is pre-configured by the admin — just open the URL and start searching immediately.

WHAT YOU NEED

- A modern web browser
- The app URL from your administrator
- An internet connection

NO SETUP REQUIRED

The API key is pre-loaded. No configuration needed on your end.

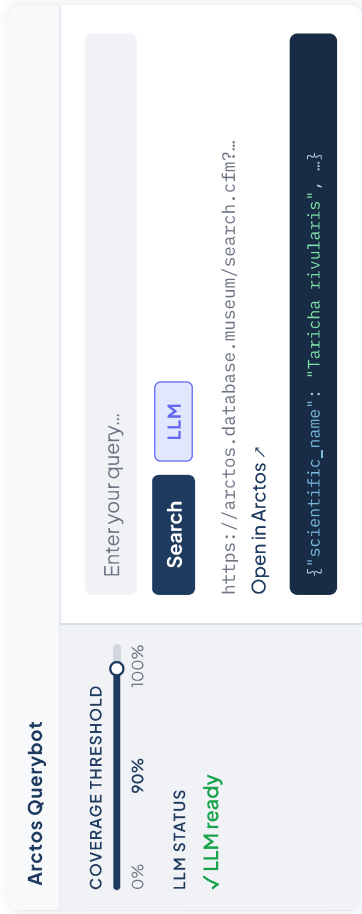
The interface

SIDEBAR

- **Coverage threshold slider** — controls the cutoff between local entity matching and AI-powered search (default: 90%)
- **LLM status** — shows whether the Gemini model is ready (✓ green when active)

MAIN AREA

- **Query input** — type your plain-English question here
- **Search button** — submits the query
- **Route badge** — LOCAL or LLM, indicating which search path was taken
- **Result URL** — click "Open in Arctos ↗" to view live records
- **Extracted fields** — the structured parameters parsed from your query



Entering a query

Type your question in the search box using plain English. No special syntax or codes are required.

EXAMPLE QUERIES

"Show me collections from MVZ:Herp"
"What specimens of Taricha rivularis are in MVZ:Herp?"
"Bird specimens collected by John Doe after 1990"
"ALMNH:Bird and APSU:Fish collections"

TIPS FOR BETTER RESULTS

- Use specific GUID prefixes (e.g., MVZ:Herp) when you know them — they route locally and return faster, exact results
- Spell scientific names correctly — they are extracted as-is by the LLM
- Include dates, collectors, and geography to filter results further

Click **Search** or press Enter to submit. Results appear below the input field.

How queries are routed

The app automatically picks one of two search strategies based on how much of your query is covered by known institutions, collections, or GUID prefixes.

LOCAL

Used when the query is dominated by known entities — institutions, collections, or GUID prefixes.

No AI call is made. The URL is built directly from matched entities. Fast and exact.

LLM

Used when the query contains fields the app doesn't recognise — taxonomy, dates, geography, or collector names.

Gemini AI extracts the relevant fields and maps them to Arctos URL parameters.

Reading your results

1 • THE ARCTOS URL

The generated search link. Click **Open in Arctos** ↗ to see matching specimen records live in the Arctos database.

2 • EXTRACTED FIELDS

The parameters the app parsed from your query, displayed as JSON. Review these to confirm the app understood your question correctly. If a field looks wrong, try rephrasing your query.

3 • ENTITY MATCHES

A collapsible panel listing the institutions, collections, and GUID prefixes the preprocessor found. Expand it to see what was matched before any AI processing.

Extracted fields

```
{
  "guid_prefix": "MVZ:Herp"
  "scientific_name": "Taricha rivularis"
}
```

> Entity matches (from preprocessor)

Thanks for the feedback!

The Extracted fields panel shows the parsed JSON and a collapsible entity-matches section below it.

Leaving feedback

After every search result, a feedback form appears at the bottom. **Feedback is entirely optional** — skip it or fill it in, your result stays visible either way.

HOW TO SUBMIT FEEDBACK

- Select  **Yes** or  **No** using the radio button to rate the result
- Optionally type a short comment (e.g. "Wrong collection returned")
- Click **Submit feedback** when ready

The page stays open after submission — the form is replaced with a **"Thanks for the feedback!"** confirmation and your result remains visible on screen.

Was this result helpful?

☒ Yes ☐ No

Optional comment

The feedback form appears below every result. Rating and comment are both optional — click Submit when ready.

Extracted fields

```
{
  "guid_prefix": "MVZ:Herp"
  "scientific_name": "Taricha rivularis"
}
```

After submitting, the confirmation replaces the form. Your result stays on screen.

Coverage threshold

The coverage threshold slider in the sidebar controls when the AI model is used. Coverage is the proportion of your query's meaningful characters that match known entities.

≥ THRESHOLD → LOCAL ROUTE

Matched entities explain most of the query. No AI call is made — fast and exact.

< THRESHOLD → LLM ROUTE

Gemini fills in unrecognised fields — slower but handles complex queries.

DEFAULT: 90%

At 90%, queries dominated by known GUID prefixes route locally. Queries with taxonomy, dates, or geography route through the LLM.

- **Lower the threshold** — more queries route through the LLM (better for complex or mixed searches)
- **Raise the threshold** — only heavily entity-matched queries bypass the LLM
- The default of 90% balances speed and accuracy for most use cases

Tips & best practices

USE SPECIFIC ENTITY CODES

Use GUID prefixes like MVZ : Help when you know them — they route locally and return faster, exact results without an AI call.

VERIFY EXTRACTED FIELDS BEFORE CLICKING

Always review the Extracted fields panel to confirm the app understood your question correctly before opening the Arctos URL.

SPELL SCIENTIFIC NAMES CORRECTLY

The LLM extracts taxon names as-is. A misspelled binomial won't match Arctos records. Double-check names before searching.

REPHRASE AND USE FEEDBACK

If the extracted fields look wrong, try rewording your query with more specific entity names. Use the feedback form to report incorrect results — it helps improve the system.

Known limitations

- **Scientific name extraction** — the preprocessor only matches institutions, collections, and GUID prefixes. Taxon names require the LLM route to be identified
- **Ambiguous abbreviations** — a short abbreviation like "MVZ" expands to all GUID prefixes for that institution rather than a specific collection
- **LLM accuracy** — the AI may occasionally misinterpret geographic terms, dates, or collector names; always verify extracted fields

- **No query history** — previous searches are not saved between sessions; each session starts fresh
- **URL validity** — the generated URL reflects extracted fields, but the Arctos search may return zero results if the parameter combination doesn't match any specimens
- Use the feedback form to report incorrect results — it helps improve the system over time